

CHM2045 Final Exam

Information in this box is provided on the exam

Constants

Measurements of Pressure $1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ Torr}$

Gas Constants $R = 0.0821 \text{ L} \cdot \text{atm} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

Avogadro Constant $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$

Temperature $0^\circ\text{C} = 273.15 \text{ K}$

Equations

$$PV = nRT \quad \frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2} \quad \frac{\text{rate}_1}{\text{rate}_2} = \sqrt{\frac{\text{molar mass}_2}{\text{molar mass}_1}} \quad q = mc_s \Delta T$$

The following information are not provided, and it's recommended that you know these.

$$c = \lambda \nu \quad E = h\nu \quad E = \frac{hc}{\lambda}$$
$$\Delta E = -2.18 \times 10^{-18} \text{ J} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$\Delta H^\circ = \sum \text{bonds broken} - \sum \text{bonds formed}$$

$$\Delta H^\circ = \sum n\Delta H_f^\circ(\text{products}) - \sum n\Delta H_f^\circ(\text{reactants})$$

$$\text{atomic mass} = \sum (\text{mass of the isotope})(\text{fraction abundance})$$

Solubility Guidelines for Ionic Compounds

- Compounds that contain NO_3^- and $\text{C}_2\text{H}_3\text{O}_2^-$, Na^+ , K^+ , Li^+ , or NH_4^+ ions are soluble.
- Most halides (Cl^- , Br^- , I^-) are soluble, *except* halides of Ag^+ , Pb^{2+} , and Hg_2^{2+} .
- Most compounds that contain the SO_4^{2-} ion are soluble, *except* when paired with Ba^{2+} , Sr^{2+} , Ca^{2+} , Pb^{2+} and Ag^+ .
- Most compounds that contain S^{2-} and OH^- ions are insoluble (will form a precipitate) except the Group 1 compounds, which are soluble. Note: when S^{2-} pairs with Ca^{2+} , Sr^{2+} and Ba^{2+} , the compounds are soluble. Also, $\text{Ba}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$ are slightly soluble.
- Most compounds that contain CO_3^{2-} and PO_4^{3-} ions are insoluble (will form a precipitate) *except* the Group 1 and NH_4^+ compounds, which are soluble.